GO

Go

Source: Notes.md

Source: https://www.youtube.com/watch?v=V-lI7AmusGs

Introduction

1. Go(Golang) was designed at Google
2. It was designed to have high performance, yet simple and easy to understand syntax
3. Go is statically typed and compiled language
4. Useful for running performant backend services; you are not going to use it to build user interfaces or entire websites.
5. Great for networking and multiprocessing

Go Usecases:

6. Go is used for cloud and network services
7. It is good for command line interfaces
8. It is good for backend web development (creating APIs, authentication services, etc.)
9. It is good for automation and DevOps.
10. It is good for utilities and standalone tools.

Interpreters and Compilers

11. When we(human) write code, we write something known as “source code”.
12. Before this code can be executed by our computer, it must first be converted into a language that machine can understand.
13. A compiler transforms our source code into something called byte code/machine code, that can be run directly by the CPU.
14. An interpreter transforms and executes source code one line at a time.
15. Javascript is an interpreted language whereas Go is a compiled language.
16. A compiler can perform a lot of optimizations on our code that allow it to run a lot faster than if we passed it to an interpreter.

Dynamic vs Static Typing

It is a property of a programming language that determines the type of a variable at run time.

17. Dynamic/Static typing refers to when and how the types of variables are decided.
18. In dynamically typed languages, variable types are decided at run time, while in statically typed languages, they are decided at compile time.
19. Javascript is a dynamically typed language whereas Go is a statically typed language.

For example, in Javascript, initially if we write var x=1 then later in the program we can change it to string/object or any other object.

If we know the type of a variable (like integer/float, etc.) before executing the code, then we can do many optimizations in our code, which is what happens in statically typed languages.